

TABLE 9.1

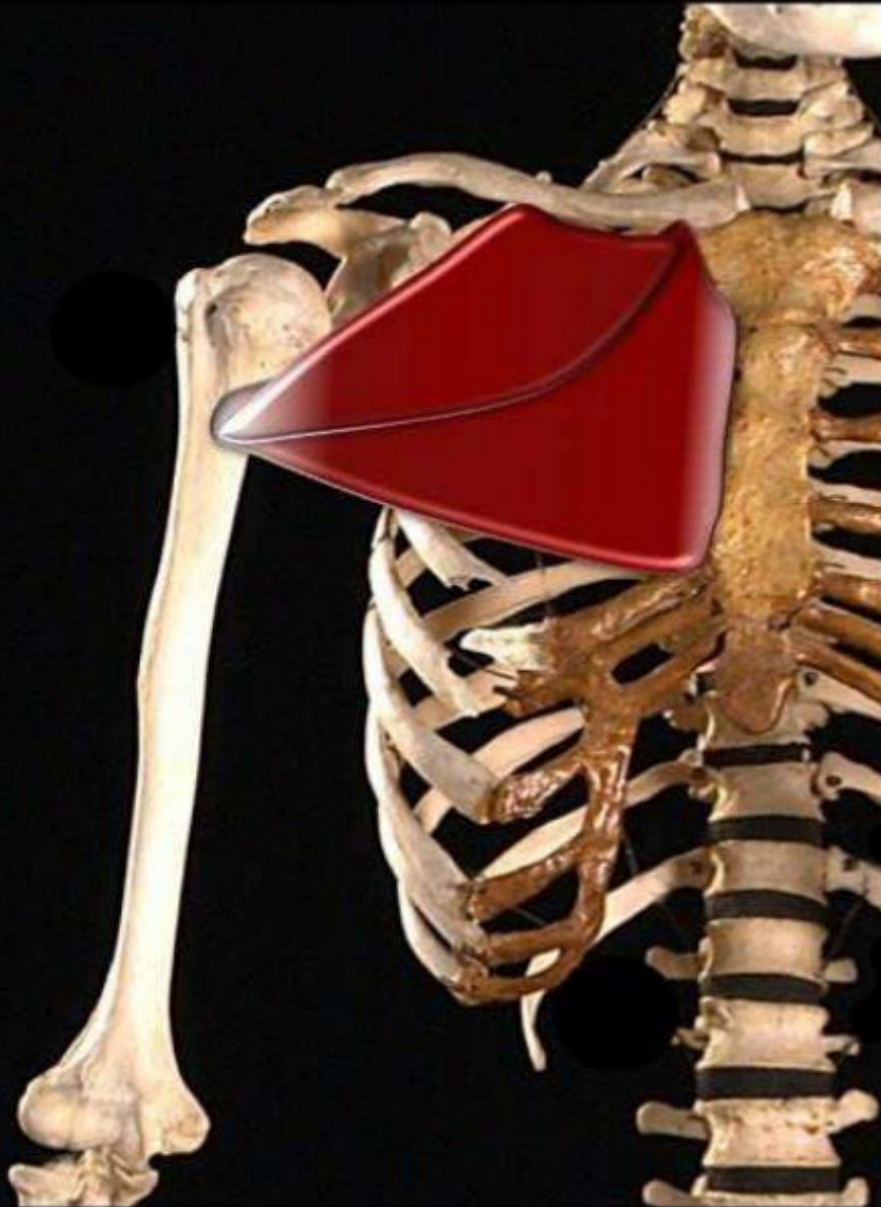
Muscles Connecting the Upper Limb to the Thoracic Wall

Muscle	Origin	Insertion	Nerve Supply	Nerve Roots ^a	Action
Pectoralis major	Clavicle, sternum, and upper six costal cartilages	Lateral lip of bicipital groove of humerus	Medial and lateral pectoral nerves from brachial plexus	C5, 6, 7, 8 ; T1	Adducts arm and rotates it medially; clavicular fibers also flex arm
Pectoralis minor	3rd, 4th, and 5th ribs	Coracoid process of scapula	Medial pectoral nerve from brachial plexus	C6, 7, 8	Depresses point of shoulder; if the scapula is fixed, it elevates the ribs of origin
Subclavius	1st costal cartilage	Clavicle	Nerve to subclavius from upper trunk of brachial plexus	C5, 6	Depresses the clavicle and steadies this bone during movements of the shoulder girdle
Serratus anterior	Upper eight ribs	Medial border and inferior angle of scapula	Long thoracic nerve	C5, 6, 7	Draws the forward anterior around the thoracic wall; rotates scapula

^aThe predominant nerve root supply is indicated by boldface type.

Anterior Chest Wall

Pectoralis Major



Origin

Clavicular head: anterior surface of medial half of clavicle; Sternocostal head: anterior surface manubrium & sternum, superior 6 costal cartilages, and aponeurosis of external oblique muscle

Insertion

Lateral lip of bicipital groove of humerus

Action

Clavicular head: flexes & adducts arm
Sternal head: adducts & internally rotates arm

Nerve Supply

Lateral and medial pectoral nerves (C5, C6, C7, C8, and T1)

Blood Supply

pectoral branch of thoracoacromial trunk

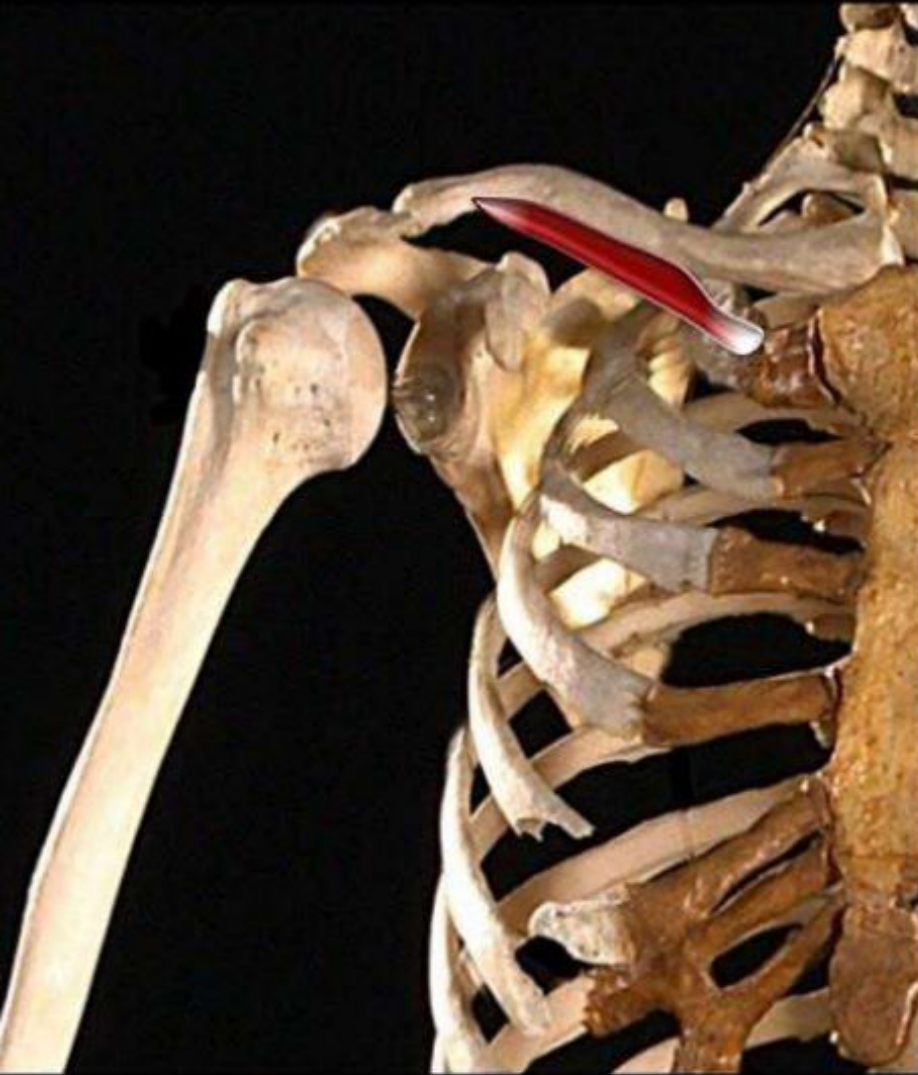
Anterior Chest Wall

Pectoralis Minor

Origin	3rd to 5th ribs near their costal cartilages
Insertion	Medial border and superior surface of coracoid process of scapula
Action	Stabilises scapula, pulls it inferiorly and anteriorly against thoracic wall
Nerve Supply	Medial pectoral nerves; (C8 and T1)
Blood Supply	Pectoral branch of the thoracoacromial trunk



Anterior Chest Wall



Subclavius

Origin	Costochondral junction 1 st rib
Insertion	Subclavian groove inferior surface middle clavicle
Action	Depresses clavicle
Nerve Supply	Nerve to subclavius C5, 6
Blood Supply	Clavicular branch of thoraco-acromial artery and suprascapular artery

Anterior Chest Wall

Serratus Anterior

Origin	Superolateral surfaces of upper 8 or 9 ribs at the side of chest
Insertion	Medial border of scapula
Action	Pulls scapula forward and upward; abducts and rotates it; stabilises medial border of scapula
Nerve Supply	Long thoracic nerve (C5, C6, C7)
Blood Supply	Circumflex scapular artery



TABLE 9.2

Muscles Connecting the Upper Limb to the Vertebral Column

Muscle	Origin	Insertion	Nerve Supply	Nerve Roots ^a	Action
Trapezius	Occipital bone, ligamentum nuchae, spine of 7th cervical vertebra, spines of all thoracic vertebrae	Upper fibers into lateral third of clavicle; middle and lower fibers into acromion and spine of scapula	Spinal part of accessory nerve (motor) and C3 and 4 (sensory)	XI cranial nerve (spinal part)	Upper fibers elevate the scapula; middle fibers pull scapula medially; lower fibers pull medial border of scapula downward
Latissimus dorsi	Iliac crest, lumbar fascia, spines of lower six thoracic vertebrae, lower three or four ribs, and inferior angle of scapula	Floor of bicipital groove of humerus	Thoracodorsal nerve	C6, 7 , 8,	Extends, adducts, and medially rotates the arm
Levator scapulae	Transverse processes of 1st four cervical vertebrae	Medial border of scapula	C3 and 4 and dorsal scapular nerve	C3, 4, 5	Raises medial border of scapula
Rhomboid minor	Ligamentum nuchae and spines of 7th cervical and 1st thoracic vertebrae	Medial border of scapula	Dorsal scapular nerve	C4 , 5	Raises medial border of scapula upward and medially
Rhomboid major	Second to 5th thoracic spines	Medial border of scapula	Dorsal scapular nerve	C4 , 5	Raises medial border of scapula upward and medially

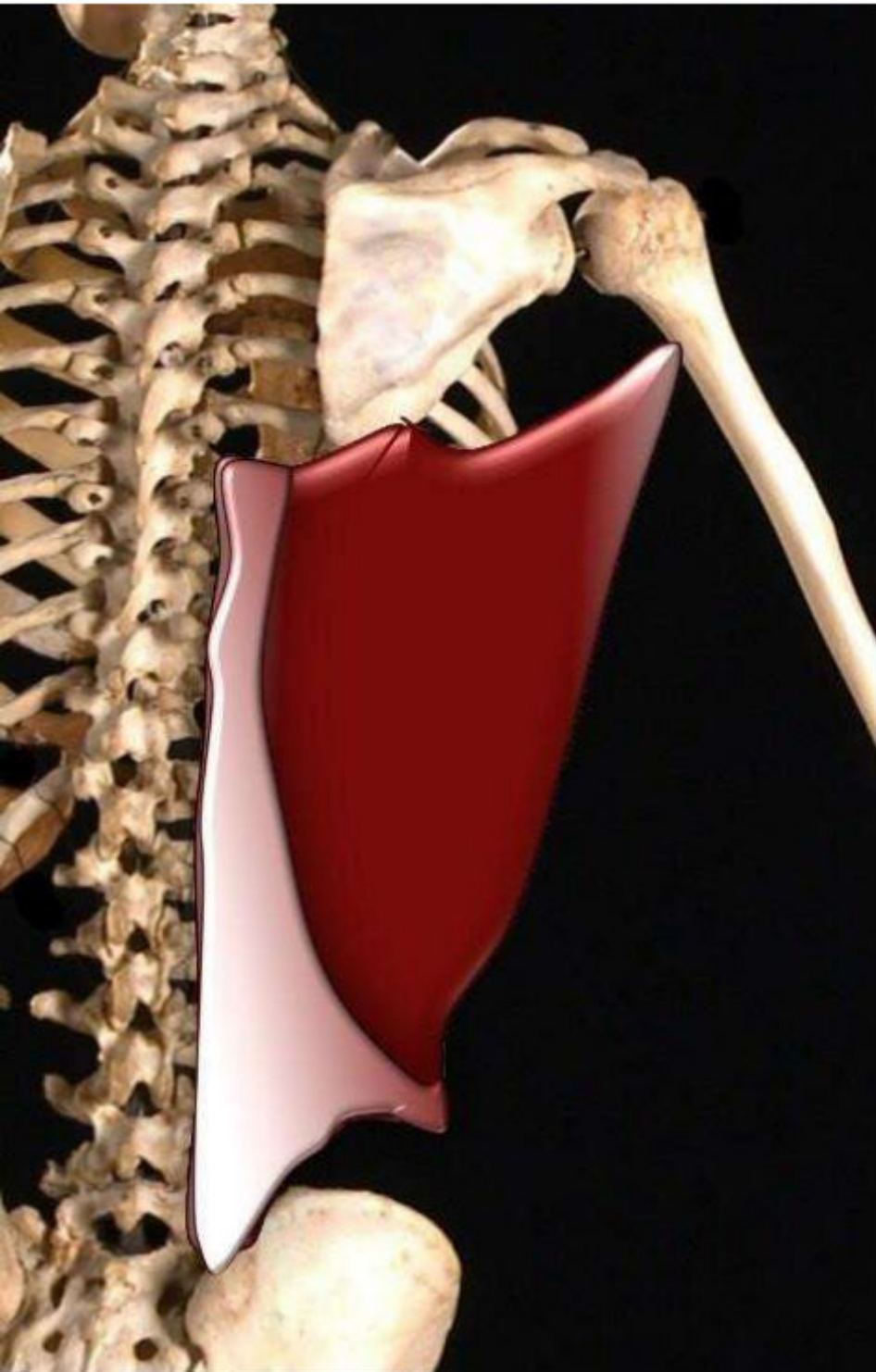
^aThe predominant nerve root supply is indicated by boldface type.

Posterior Chest Wall



Trapezius

Origin	Medial third of superior nuchal line; external occipital protuberance, nuchal ligament, and spinous processes of thoracic vertebrae
Insertion	Lateral third of clavicle, acromion, and spine of scapula
Action	superior fibers elevate, middle fibers retract, and inferior fibers depress scapula
Nerve Supply	Spinal root of accessory nerve (CN XI) (motor) and cervical nerves (C3 and C4) (pain and proprioception)
Blood Supply	Transverse cervical artery



Posterior Chest Wall

Latissimus Dorsi

Origin	Spinous processes of inferior 6 thoracic vertebrae, thoracolumbar fascia, iliac crest, and inferior 3 or 4 ribs, inferior angle of scapula
Insertion	Floor of intertubercular groove of humerus
Action	Extends, adducts, and internally rotates humerus; pulls body up toward arms during climbing
Nerve Supply	Thoracodorsal nerve (C6, C7, and C8)
Blood Supply	Thoracodorsal artery

Posterior Chest Wall



Levator Scapulae

Origin	Posterior tubercles of transverse processes of C1 - C4 vertebrae
Insertion	Superior part of medial border of scapula
Action	Elevates scapula and tilts glenoid inferiorly by rotating scapula
Nerve Supply	Dorsal scapular (C5) and cervical (C3 and C4) nerves
Blood Supply	Dorsal scapular artery

Posterior Chest Wall

Rhomboid Minor



Origin	Nuchal ligament and spinous processes of C7 and T1 vertebrae;
Insertion	Medial border of scapula adjacent to spine
Action	Retracts and rotates scapula to move glenoid inferiorly; fixes scapula to thoracic wall
Nerve Supply	Dorsal scapular nerve (C4 and C5)
Blood Supply	Dorsal scapular artery

Posterior Chest Wall

Rhomboid Major



Origin	Spinous processes of T2 - T5 vertebrae
Insertion	Medial border of scapula
Action	Retracts and rotates scapula to move glenoid inferiorly; fixes scapula to thoracic wall
Nerve Supply	Dorsal scapular nerve (C4 and C5)
Blood Supply	Dorsal scapular artery

TABLE 9.3

Muscles Connecting the Scapula to the Humerus

Muscle	Origin	Insertion	Nerve Supply	Nerve Roots ^a	Action
Deltoid	Lateral third of clavicle, acromion, spine of scapula	Middle of lateral surface of shaft of humerus	Axillary nerve	C5 , 6	Abducts arm; anterior fibers flex and medially rotate arm; posterior fibers extend and laterally rotate arm
Supraspinatus	Supraspinous fossa of scapula	Greater tuberosity of humerus; capsule of shoulder joint	Suprascapular nerve	C4, 5 , 6	Abducts arm and stabilizes shoulder joint
Infraspinatus	Infraspinous fossa of scapula	Greater tuberosity of humerus; capsule of shoulder joint	Suprascapular nerve	(C4), 5 , 6	Laterally rotates arm and stabilizes shoulder joint
Teres major	Lower third of lateral border of scapula	Medial lip of bicipital groove of humerus	Lower subscapular nerve	C6 , 7	Medially rotates and adducts arm and stabilizes shoulder joint
Teres minor	Upper two thirds of lateral border of scapula	Greater tuberosity of humerus; capsule of shoulder joint	Axillary nerve	(C4), C5 , 6	Laterally rotates arm and stabilizes shoulder joint
Subscapularis	Subscapular fossa	Lesser tuberosity of humerus	Upper and lower subscapular nerves	C5, 6 , 7	Medially rotates arm and stabilizes shoulder joint

^aThe predominant nerve root supply is indicated by boldface type.

Posterior Chest Wall



Deltoid

Origin	Lateral third of clavicle, acromion, and spine of scapula
Insertion	Deltoid tuberosity of humerus
Action	Anterior part: flexes and internally rotates arm; Middle part: abducts arm; Posterior part: extends and externally rotates arm
Nerve Supply	Axillary nerve (C5 and C6)
Blood Supply	Deltoid branch of thoracoacromial artery

Posterior Chest Wall



Supraspinatus

Origin	Supraspinous fossa of scapula
Insertion	Superior facet on greater tuberosity of humerus
Action	Abduction of arm
Nerve Supply	Suprascapular nerve (C4, C5 and C6)
Blood Supply	Suprascapular artery

Posterior Chest Wall



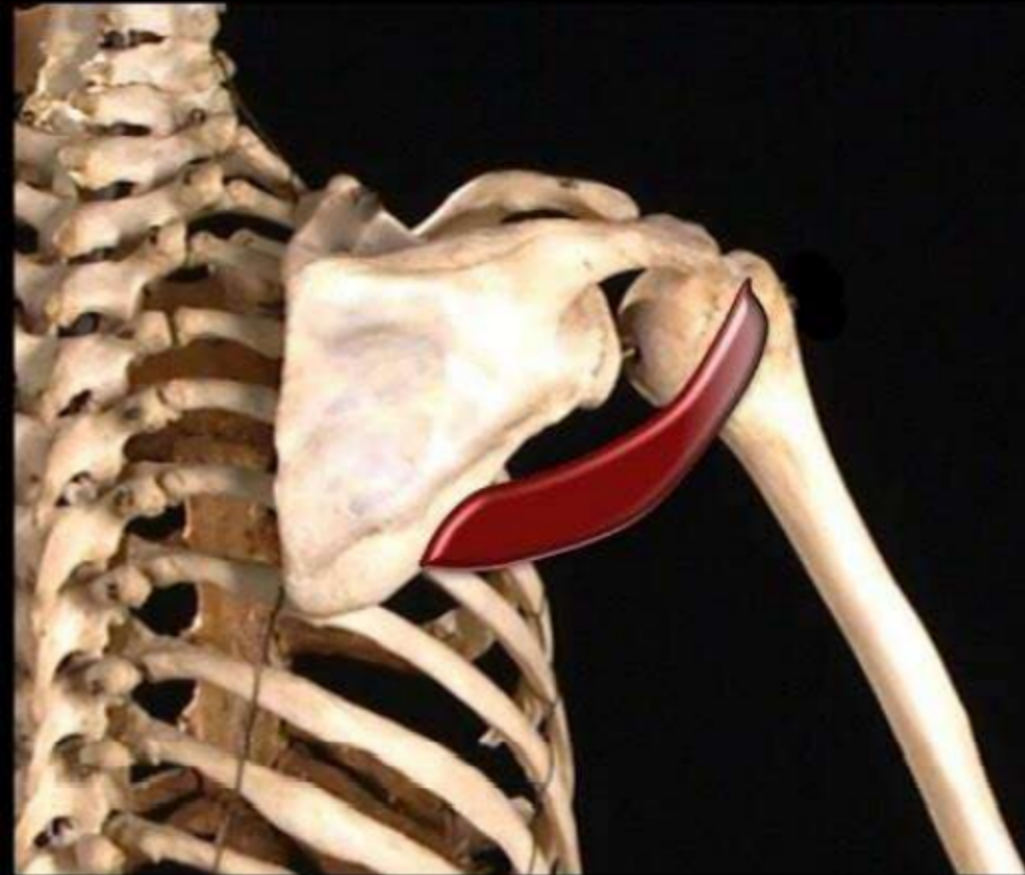
Teres Major

Origin	Dorsal surface of inferior angle of scapula
Insertion	Medial lip of intertubercular groove of humerus
Action	Adducts and internally rotates arm
Nerve Supply	Lower subscapular nerve (C6 and C7)
Blood Supply	Subscapular and circumflex scapular arteries

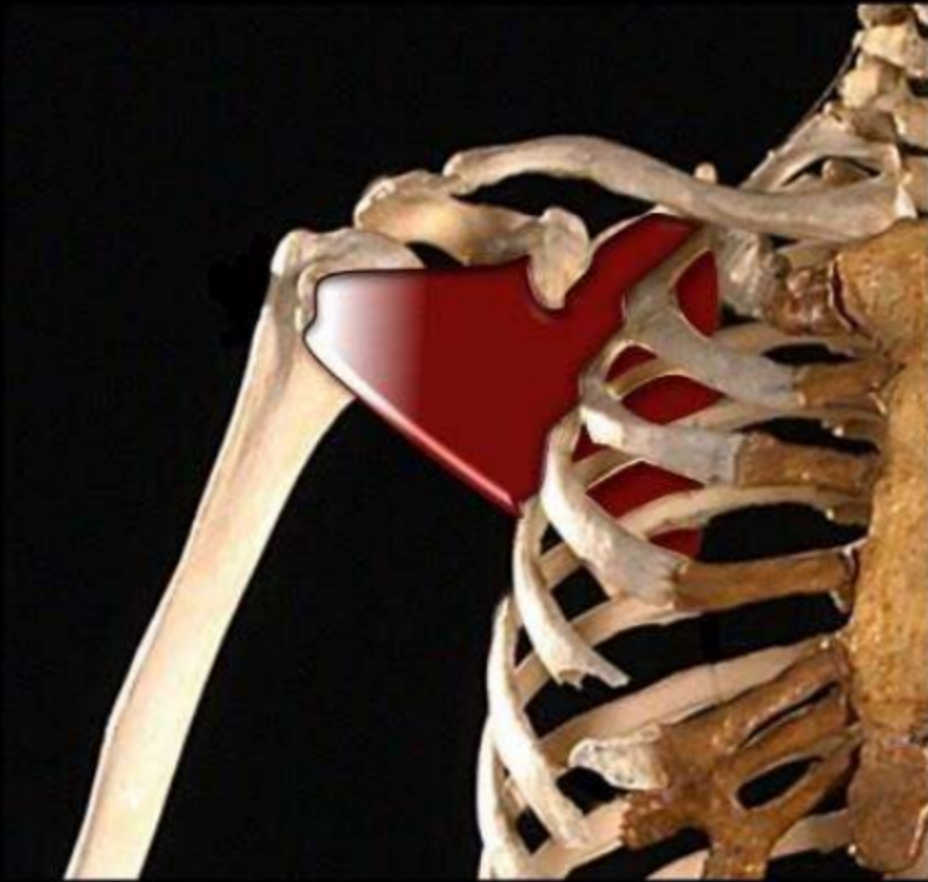
Posterior Chest Wall

Teres Minor

Origin	Superior part of lateral border of scapula
Insertion	Inferior facet on greater tuberosity of humerus
Action	External rotation of the arm
Nerve Supply	Axillary nerve (C5 and C6)
Blood Supply	Subscapular and circumflex scapular arteries



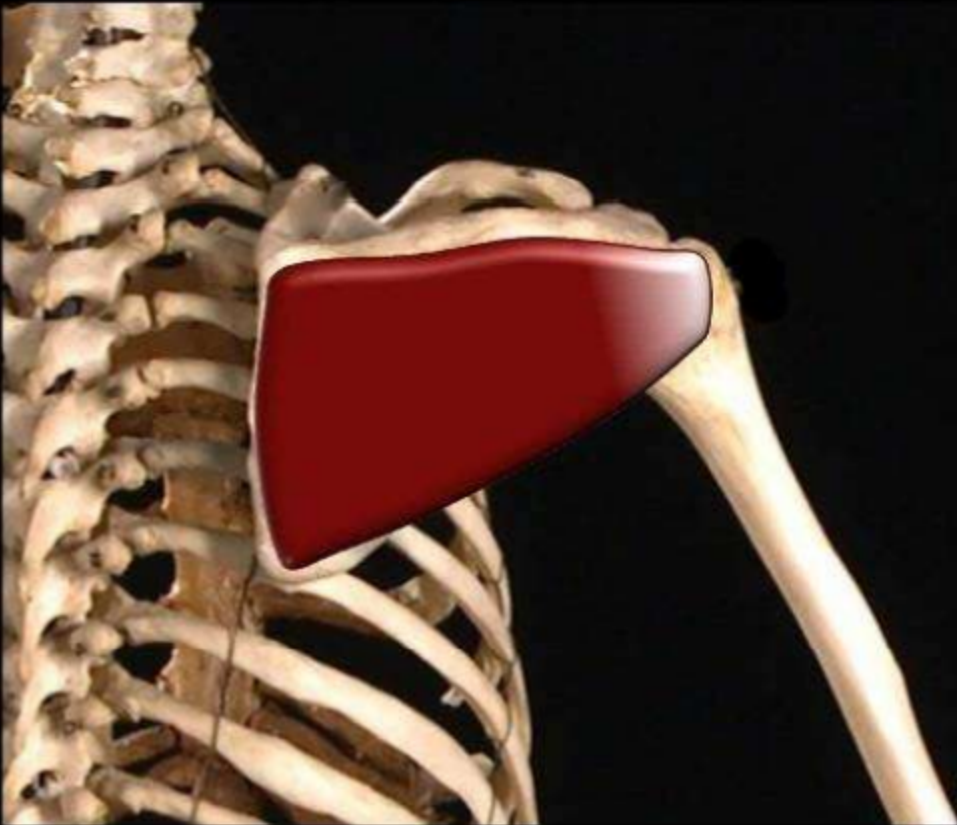
Anterior Chest Wall



Subscapularis

Origin	Subscapular fossa of scapula
Insertion	Lesser tuberosity of humerus
Action	Internal rotation and adduction of the arm
Nerve Supply	Upper and lower subscapular nerves (C5, C6 and C7)
Blood Supply	Subscapular artery

Posterior Chest Wall



Infraspinatus

Origin	Infraspinous fossa of scapula
Insertion	Middle facet on greater tuberosity of humerus
Action	Externally rotates arm
Nerve Supply	Suprascapular nerve (C5 and C6)
Blood Supply	Suprascapular and circumflex scapular arteries